



GOSFORD HIGH SCHOOL

EXTENSION 1 MATHEMATICS

HSC Course Assessment Task 1

December 2010

Special Instructions

- Students are to hand in their papers in three separate bundles.
- Students must start PART B and PART C on a new page.

General Instructions

Time Allowed: 60 minutes plus 5 minutes reading time.

- Attempt all questions.
- Approved calculators may be used.
- Write using blue or black pen.
- Full marks may not be awarded where necessary working is not shown.

PART A: ITERATIVE METHODS.

Question 1:

(a) Show that $f(x) = 2x^3 + 2x - 1$ has a zero between $x=0$ and $x=1$. (2)

(b) By considering $f'(x)$, explain why this is the only root of $f(x) = 0$. (2)

(c) Taking $x=0$ as a first approximation to this root, use Newtons Method twice to find an approximation to two decimal places. (3)

Question 2:

(a) Show that $f(x) = x^4 + 4x^3 + 8x - 4$ has a zero α between $x=0$ and $x=1$. (2)

(b) Use the method of halving the interval to determine whether α is closer to $x=0$ or $x=1$. (2)

(c) Given that $x^4 + 4x^3 + 8x - 4 = (x^2 + 2)(x^2 + 4x - 2)$, show that $f(x) = 0$ has only two real roots and find the value of α correct to three decimal places. (3)

PART B: MATHEMATICAL INDUCTION.

(Start a new page)

Question 1:

(a) Prove by induction that:

$$1.2 + 3.4 + 5.6 + \dots + (2n-1)(2n) = \frac{n}{3} (n+1) (4n-1)$$

for all positive integers n . (6)

(b) Prove by induction that:

$$2^n - 1 \text{ is divisible by 3 for all even positive integers.} \quad (5)$$

PART C: PARAMETRIC TREATMENT OF THE PARABOLA.

(Start a new page)

Question1:

(a) Show that the points defined by the parametric equations $x=2\cos t$ and $y=\cos 2t$ lie on a parabolic arc. (3)

(b) Sketch the arc clearly showing its end points, the focus and the directrix of this parabola. (2)

Question 2:

(a) The parametric equations of a parabola are $x=6t$ and $y=3t^2$. Find the Cartesian equation of the parabola. (2)

(b) Hence show that the equation of the normal at P , where $t=p$, is

$$x+py=3p^3+6p. \quad (4)$$

Question 3:

$P(2ap, ap^2)$ is a variable point on the parabola $x^2 = 4ay$ whose focus is S . The line through S , perpendicular to the tangent at P , meets the tangent at L .

(a) Show that the equation of the tangent at P is $y = px - ap^2$. (3)

(b) Show that the equation of SL is $x + py = ap$. (2)

(c) Determine the locus of L . (2)

Part A

Q1/

$$a) f(0) = 2(0)^3 + 2(0) - 1$$

$$= -1$$

$$f(1) = 2(1)^3 + 2(1) - 1$$

$$= 3$$

Since $f(x)$ is continuous

* Since $f(0)$ & $f(1)$ are opposite in sign there is a zero of $f(x)$ between $x=0$ & $x=1$

$$b) f(x) = 2x^3 + 2x - 1$$

$$f'(x) = 6x^2 + 2$$

$\therefore f'(x) > 0$ for all x and hence $f(x)$ is a monotonically increasing function. & continuous
 $\therefore$ There is only one root of $f(x) = 0$

$$c) a_2 = a_1 - \frac{f(a_1)}{f'(a_1)}$$

$$= 0 - \frac{-1}{2}$$

$$= \frac{1}{2}$$

$$a_3 = a_2 - \frac{f(a_2)}{f'(a_2)}$$

$$= \frac{1}{2} - \frac{\frac{1}{4}}{\frac{5}{2}}$$

$$\approx 0.43 \text{ (2 d.p.)}$$

Q2/

Note $f(x)$ is continuous

$$a) f(0) = (0)^4 + 4(0)^3 + 8(0) - 4$$

$$= -4$$

$$f(1) = (1)^4 + 4(1)^3 + 8(1) - 4$$

$$= 9$$

Since $f(0)$ & $f(1)$ are opposite in sign, α lies between $x=0$ & $x=1$.

$$b) f(0.5) = (0.5)^4 + 4(0.5)^3 + 8(0.5) - 4$$

$$= 0.5625$$

Since $f(0)$ & $f(0.5)$ are opposite in sign α lies between $x=0$ & $x=0.5$

$\therefore \alpha$ is closer to 0 than 1.

$$c) \text{ If } f(x) = 0$$

$$x^2 + 2 = 0$$

No real solⁿ

$$\text{or } x^2 + 4x - 2 = 0$$

$$x^2 + 4x + 4 = 2 + 4$$

$$(x+2)^2 = 6$$

$$x+2 = \pm\sqrt{6}$$

$$x = -2 \pm \sqrt{6}$$

$\therefore f(x)$ has only two real roots

Now $-2 + \sqrt{6}$ is between 0 & 1

$$\therefore \alpha \approx 0.449 \text{ (3 d.p.)}$$

Part B

Q1

$$a) 1 \cdot 2 + 3 \cdot 4 + \dots + (2n-1)2n = \frac{n}{3}(n+1)(4n-1)$$

Prove true for $n=1$

$$LHS = 1 \cdot 2$$

$$= 2$$

$$RHS = \frac{1}{3}(1+1)(4-1)$$

$$= 2$$

$\therefore$ true for $n=1$.

Assume true for $n=k$

$$\text{i.e. } 1 \cdot 2 + 3 \cdot 4 + \dots + (2k-1)2k = \frac{k}{3}(k+1)(4k-1)$$

Prove true for $n=k+1$

$$\text{i.e. } 1 \cdot 2 + 3 \cdot 4 + \dots + (2k-1)2k + (2k+1)(2k+2) = \frac{k+1}{3}(k+2)(4k+3)$$

$$LHS = \frac{k}{3}(k+1)(4k-1) + (2k+1)(2k+2)$$

$$= \frac{k}{3}(k+1)(4k-1) + 2(2k+1)(k+1)$$

$$= \frac{k+1}{3} [k(4k-1) + 6(2k+1)]$$

$$= \frac{k+1}{3} (4k^2 + 11k + 6)$$

$$= \frac{k+1}{3} (4k+3)(k+2)$$

$$= RHS$$

$\therefore$ If the statement is true for $n=k$, it is true for $n=k+1$. Since it is true for $n=1$, by induction the statement is true for all positive integers n .

b) $2^n - 1$ is divisible by 3 for all even positive integers

Prove true for $n=2$

$$2^2 - 1 = 3$$

which is divisible by 3.

Assume true for $n=k$, where k is a positive even integer.

Let $2^k - 1 = 3Q$ for some integer Q

Prove true for $n=k+2$

$$\text{Now } 2^{k+2} - 1 = 2^2(2^k - 1) + 3$$

$$= 4 \times 3Q + 3$$

$$= 3(4Q + 1)$$

which is divisible by 3.

$\therefore$ If the statement is true for $n=k$, it is true for $n=k+2$. Since it is true for $n=2$, by induction it is true for all positive even integers n .

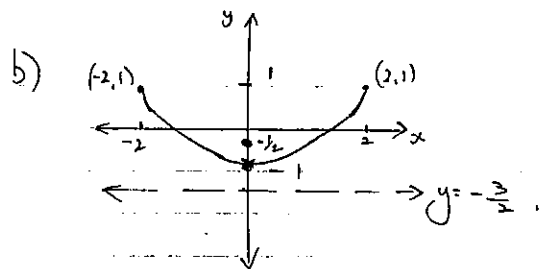
PART C

Q1

a) $x = 2 \cos t$, $y = \cos 2t$
 Since $-1 \leq \cos t \leq 1$
 $-2 \leq 2 \cos t \leq 2$
 $\therefore -2 \leq x \leq 2$

Also if $y = \cos 2t$
 $y = 2 \cos^2 t - 1$
 $y = 2 \left(\frac{x}{2}\right)^2 - 1$
 $y = \frac{x^2}{2} - 1$
 $2y = x^2 - 2$
 $x^2 = 2y + 2$
 $x^2 = 4 \times \frac{1}{2} (y + 1)$

$\therefore$ the eqn represents a parabolic arc with vertex at $(0, -1)$ & focal length $\frac{1}{2}$ unit for $-2 \leq x \leq 2$.



N.B. If $x = -2$, $y = 1$
 $x = 2$, $y = 1$

Q2

a) $x = 6t$, $y = 3t^2$
 If $x = 6t$
 $t = \frac{x}{6}$

$\therefore y = 3 \left(\frac{x}{6}\right)^2$
 $y = \frac{3x^2}{36}$
 $y = \frac{x^2}{12}$

or $x^2 = 12y$

b) If $y = \frac{x^2}{12}$

$y' = \frac{2x}{12}$

$= \frac{x}{6}$

When $x = 6t$, $y' = \frac{6t}{6} = t$

$\therefore$ the gradient of the normal is $-\frac{1}{t}$

Hence the eqn of the normal is :

$y - 3t^2 = -\frac{1}{t}(x - 6t)$

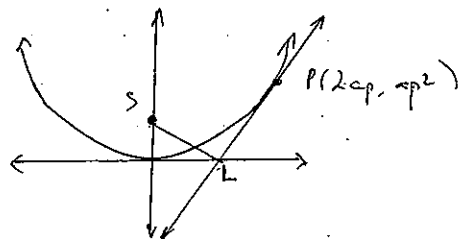
$ty - 3t^3 = -x + 6t$

i.e. $x + ty = 3t^3 + 6t$

Now when $t = p$

$x + py = 3p^3 + 6p$

Q3



2) S is the point $(0, a)$

$$\text{If } x^2 = 4ay, \quad y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a}$$

$$= \frac{x}{2a}$$

$$\text{When } x = 2ap, \quad y' = \frac{2ap}{2a}$$

$$y' = p$$

$\therefore$ the eqn of the tangent is:

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

— (1)

b) m of SL is $-\frac{1}{p}$

$\therefore$ the eqn of SL is:

$$y = -\frac{1}{p}x + a$$

— (2)

$$\text{i.e. } py = -x + ap$$

$$x + py = ap$$

b) Solving (1) & (2) simultaneously

$$-\frac{1}{p}x + a = px - ap^2$$

$$-x + ap = p^2x - ap^3$$

$$\therefore ap^3 + ap = p^2x + x$$

$$\text{Hence } x(p^2 + 1) = ap(p^2 + 1) \quad \text{since } p^2 + 1 \neq 0$$

$$\therefore x = ap$$

$\therefore$ Substituting into (2)

$$y = -\frac{1}{p} \cdot ap + a$$

$$y = -a + a$$

$$y = 0$$

$\therefore$ the locus is the line $y = 0$ (or the x-axis).